Semester II Examinations 2017/2018

Exam Code(s)	3BA1, 4BA4, 3BS9, 3MR31, 4BS2, 4BME1, 1CSD1
Exam(s)	Third & Fourth Arts; Third & Fourth Science; Fourth Maths and Education; MSc Data Analytics
Module(s)	Applied Statistics 2
Module Code(s)	ST312
Paper No	1
Repeat Paper	
External Examiner	Prof. S. Wilson
Internal Examiner	Prof. J. P. Hinde

Instructions:

Answer ALL 5 Questions.

Duration	2 Hours
No. of Pages	12 Pages
Department(s)	Statistics
Course Co-ordinators(s)	

Requirements:

Release to Library:	Yes
Statistical Tables/ Log Tables	Yes
Other Materials	Standard (non-programmable) calculator is allowed

Question 1: [Total marks 15]

Hay fever is a problem for many individuals and good, effective treatments do not really exist. In a study, two new treatments, from different manufacturers, are compared with a placebo using 18 volunteers in the hay-fever season. For convenience in one week the physicians give treatment A to six volunteers and the placebo to three; in the following week they give treatment B to six more volunteers and the placebo to three more. They measure the respiratory function of each volunteer before treatment and three hours after treatment.

(a) Is this an observational or experimental study? Explain your answer. *(2 marks)*

(b) Identify the response and explanatory variables, the treatments, and the experimental units. *(4 marks)*

(c) Are the results generalisable from the sample of patients to the population of hay-fever sufferers? *(2 marks)*

(d) How should the researchers assign the subjects to the two groups? Why? Explain the advantages of your proposed approach. *(3 marks)*

(e) Does this experiment give a fair comparison between the two new treatments? If not, why not? *(2 marks)*

(f) How could this study design be improved? *(2 marks)*

Question 2: [Total marks 15]

An experiment was conducted to compare two protective dyes for metal, both with each other and with 'no dye'. Eight braided metal cords were broken into three pieces. The three pieces of each cord were randomly allocated to the three treatments. After the dyes had been applied, the cords were left to weather for a fixed time, then their strengths were measured. The table below shows the strengths as a percentage of the nominal strength specification, with the values given in a systematic order rather than the random order of the experiment.

	Cord								
	1	2	3	4	5	6	7	8	Total
no dye	102.4	93.7	97.4	96.1	102.5	87.8	102.6	95.2	777.7
dye A	108.5	92.3	93.1	106.9	92.0	95.5	108.4	94.6	791.3
dye B	106.8	96.7	100.6	101.9	103.3	94.9	106.5	101.2	808.9
Cord Total	317.7	282.7	291.1	304.9	297.8	278.2	317.5	291.0	

(a) List the factor of interest and any nuisance factors. *(2 marks)*

(b) What are the experimental units? *(1 mark)*

(c) Are there blocks in this design? If so, identify them. *(2 marks)*

(d) Why did the experimenters conduct the experiment in this way? *(2 marks)*

(e) Has their approach been justified by the data? *(2 marks)*

(f) Describe how a randomization test could be used to see if the dyes have any effect on strength, taking account of the possible between cord variation. State the null and alternative hypotheses being tested and describe (briefly) how such a test would be implemented including how the associated p-value would be calculated. What would be the advantage of doing this rather than a standard ANOVA F-test?
(**No calculations are required.**) *(6 marks)*

Question 3: [Total marks 35]

As part of a quality exercise at a manufacturing plant, five machine components were chosen at random from the output of four different workers on two different days and the weight (in grams) was used as the quality measure.

(a) An initial analysis using analysis of variance looked to see if there was any difference between the two days. Use the output below to test the hypothesis that there was no difference between the two days. Clearly specify your null and alternative hypotheses, calculate an appropriate F-statistic and use the tables at the end of this paper to perform a test at the 5% level.

```
One-way ANOVA: weight versus day

Source  DF       SS      MS  F-Value  P-Value
day      1    6.084   6.084     ****     ****
Error   38   97.776   2.573
Total   39  103.860
```

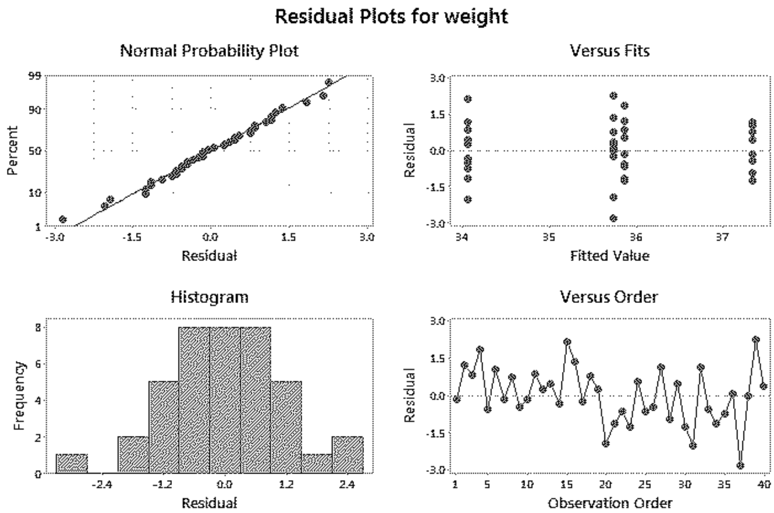
An alternative would have been to use a two-sample t-test; what would have been the value of the t-test statistic? *(6 marks)*

(b) Next they looked to see if there was any difference between the workers. Using the Minitab output below, test the hypothesis that there are no differences between the workers. Again, clearly specify your null and alternative hypotheses, perform an F-test at the 5% level, and state your conclusions. *(4 marks)*

```
Analysis of Variance

Source  DF      SS      MS  F-Value
worker   3   54.62  18.207    13.31
Error   36   49.24   1.368
Total   39  103.86

worker   N    Mean  StDev       95% CI
1       10  35.860  1.057  (35.110, 36.610)
2       10  37.350  0.824  (36.600, 38.100)
3       10  34.050  1.222  (33.300, 34.800)
4       10  35.740  1.477  (34.990, 36.490)

Pooled StDev = 1.16950
```

Question continues . . .

(c) Comment on the validity of the assumptions for this ANOVA based on the residual plots below. If you have reservations on the validity, suggest what you might do to improve matters. *(3 marks)*



(d) Following on from the above ANOVA why does it make sense to perform a multiple comparisons procedure?
Use the following output to perform multiple comparisons. What are the *family-wise* and *individual* type I error rates of your procedure? Group the different workers accordingly. *(4 marks)*

```
Tukey 95% Simultaneous Confidence Intervals for Differences of Means

Difference  Difference       SE of
of Levels     of Means  Difference       95% CI
2 - 1            1.490       0.523  ( 0.081,  2.899)
3 - 1           -1.810       0.523  (-3.219, -0.401)
4 - 1           -0.120       0.523  (-1.529,  1.289)
3 - 2           -3.300       0.523  (-4.709, -1.891)
4 - 2           -1.610       0.523  (-3.019, -0.201)
4 - 3            1.690       0.523  ( 0.281,  3.099)

Individual confidence level = 98.93%
```

(e) The first two workers are relatively new, while workers 3 and 4 are more experienced. Define a suitable contrast to compare the population mean of workers 1 and 2 with that for workers 3 and 4. Specify the null and alternative hypotheses being tested, obtain an estimate for the contrast statistic, calculate its standard error and the degrees of freedom for the test. Perform the test at the 5% significance level. *(6 marks)*

Question continues . . .

(f) To consider both factors, day and worker, a two-way ANOVA was performed with the output given below. What do you conclude? Relate your conclusions to your previous findings, in particular in part (a). *(4 marks)*

```
ANOVA: weight versus day, worker

Source  DF       SS      MS      F      P
day      1    6.084   6.084   4.93  0.033
worker   3   54.622  18.207  14.77  0.000
Error   35   43.154   1.233
Total   39  103.860
```

(g) The model in the above analysis is just a main effects model. The tables below give the cell means (based on 5 observations per cell) and a partial table of interaction effects. Complete the entries in the interaction effects table.
If there were no interaction, what would be the values in this table? *(3 marks)*

	Worker				Row
Cell means	1	2	3	4	Mean
Day 1	36.50	37.56	34.72	35.78	36.14
Day 2	35.22	37.14	33.38	35.70	35.36
Column Mean	35.86	37.35	34.05	35.74	35.75

	Worker			
Interaction Effect	1	2	3	4
Day 1	0.25	**A**	0.28	−0.35
Day 2	−0.25	**B**	−0.28	0.35

(h) Calculate the Interaction Sum of Squares (remembering that there are 5 observations per cell) and its degrees of freedom. Hence, extend the two-way ANOVA table in (f) to include the interaction term, calculating the new Error SS, Error MS and effect F-statistics. Test the significance of the interaction. What do you conclude? *(5 marks)*

Question 4: [Total marks 10]

Dietary fibre is a healthy component of many vegetables and grains. Heavily processed foods are low in fibre, so it is often added to such foods to make them more nutritious. In this study subjects were given one of two types of cracker before a meal and then asked shortly after to report any bloating that they experienced. Participants were randomly assigned to two types of cracker, one fibre-enriched with bran and the other standard with no fibre additive. 12 participants were randomly assigned to each cracker type. The results are given in the following 2×2 table.

Cracker	Bloating		
Type	None	Some	Total
None	6	6	12
Bran	2	10	12
Total	8	16	20

Compare bloating from the two cracker types using the following measures and state what the relevant null and alternative hypotheses would be. You do not need to perform the test, simply calculate the test statistics.

(i) Equality of proportions;

(ii) Relative Risk;

(iii) Odds-ratio.

(6 marks)

For small studies, such as this, standard large-sample results based on the normal distribution can be unreliable. Fisher's exact test is designed to deal with this problem and calculates a randomization p-value based on the observed table and table(s) with the same marginal totals that show stronger evidence of a treatment effect. List the table(s) that are more extreme than the above table. *(4 marks)*

Question 5: [Total marks 25]

The question relates to an historical study of chronic bronchitis in Cardiff. The data consist of observations for a sample of 212 men and include whether he smoked, `smok` ($1 =$ yes; $0 =$ no), a measure of smoke pollution in the locality of the respondent's home, `poll`, (obtained from air pollution monitoring stations across the city), and a binary indicator `r` taking the value 1 if the respondent suffered from chronic bronchitis and 0 if he did not.

In the following we are interested in modelling the probability of chronic bronchitis using logistic regression models.

The output below is from a model exploring the relationship between the binary variable `r` and the `poll` level.

Fit 1: `Logistic Regression`

```
Coefficients
Term         Coef  SE Coef   Z-Value  P-Value  Odds Ratio       95% CI
Constant    -8.66     2.59     -3.34    0.001
poll       0.1248   0.0434      2.88    0.004      1.1329  (1.0405, 1.2335)

Deviance Table
Source       DF  Seq Dev
Regression    1    8.519
  poll        1    8.519
Error       210  213.259
Total       211  221.778
```

(a) Write down the fitted model in both the logit and probability forms of the logistic regression model. *(3 marks)*

(b) Interpret the estimated slope for `poll`. *(2 marks)*

(c) Is there a statistically significant relationship between chronic bronchitis and pollution (`poll`)? State the null and alternative hypotheses, the Wald statistic, associated p-value, and your conclusions.
What is another alternative way of testing this relationship? *(5 marks)*

(d) Use the fitted model to predict the probability of chronic bronchitis at a pollution level of 60. *(2 marks)*

(e) Epidemiologists often summarise the risk of environmental factors in terms of LD_{50}, the (lethal) exposure level required to give a 50% disease rate (i.e. a 0.5 probability of chronic bronchitis). Use the fitted model to estimate the LD_{50}. *(3 marks)*

(f) Calculate a 95% confidence interval for the slope parameter and explain how this relates to the reported interval for the *Odds Ratio*. *(3 marks)*

Question continues . . .

The above analysis ignores possible differences between smokers and non-smokers. To explore this two more models were fitted as shown below.

Fit 2: Logistic Regression Estimates

Term	Coef	SE Coef	95% CI	Z-Value	Odds Ratio	95% CI
Constant	-10.33	2.72	(-15.66, -5.00)	-3.80		
poll	0.1400	0.0442	(0.0533, 0.2266)	3.17	1.1502	(1.0548, 1.2544)
smok						
1	0.999	0.458	(0.101, 1.897)	2.18	2.7156	(1.1062, 6.6659)

Deviances:

Source	DF	Seq Dev
Error	209	207.796
Total	211	221.778

Fit 3: Logistic Regression Table
Coefficients

Term	Coef	SE Coef	95% CI	Z-Value
Constant	-12.83	7.64	(-27.80, 2.14)	-1.68
poll	0.181	0.125	(-0.064, 0.425)	1.45
smok				
1	3.86	8.14	(-12.09, 19.80)	0.47
poll*smok				
1	-0.047	0.133	(-0.309, 0.214)	-0.35

Deviances:

Source	DF	Seq Dev
Error	208	207.670
Total	211	221.778

(i) Use the output (specifically the deviance values, `Dev`) from the three model fits to produce an analysis of deviance table for the effects of `poll`, `smok` and their interaction.
What do you conclude from this table? Justify you conclusion with a formal significance test. *(5 marks)*

(ii) Use the interaction model estimates to calculate the odds ratio for a one-unit increase in pollution for smokers and non-smokers. *(2 marks)*

Table entry for p and C is the point t^* with probability p lying above it and probability C lying between $-t^*$ and t^*.

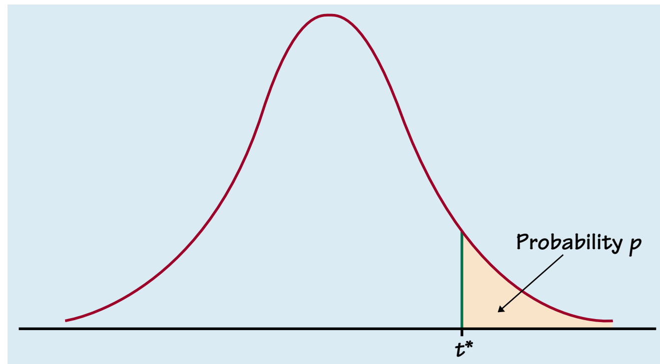
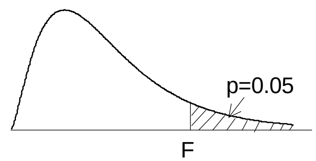


TABLE D *t* distribution critical values

	Tail probability p											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	0.674	0.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291
	50%	60%	70%	80%	90%	95%	96%	98%	99%	99.5%	99.8%	99.9%
	Confidence level C											

F distribution for p = 0.05

For various pairs of degress of freedom (v_1,v_2), the tabled entries represent the critical values of F above which the proportion p = 0.05 of the distribution falls. (Example, for df = 3,8 and F = 2.92 is exceeded by p = 0.05 of the population.



p = 0.05

	Degrees of freedom for the numerator v_1																	
v_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	120	∞
1	161.446	199.499	215.707	224.583	230.160	233.988	236.767	238.884	240.543	241.882	243.905	245.949	248.016	250.096	251.144	252.196	253.254	254.317
2	18.513	19.000	19.164	19.247	19.296	19.329	19.353	19.371	19.385	19.396	19.412	19.429	19.446	19.463	19.471	19.479	19.487	19.496
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.703	8.660	8.617	8.594	8.572	8.549	8.526
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.746	5.717	5.688	5.658	5.628
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.496	4.464	4.431	4.398	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.808	3.774	3.740	3.705	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.376	3.340	3.304	3.267	3.230
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.079	3.043	3.005	2.967	2.928
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.864	2.826	2.787	2.748	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.700	2.661	2.621	2.580	2.538
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.570	2.531	2.490	2.448	2.404
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.466	2.426	2.384	2.341	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.380	2.339	2.297	2.252	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.308	2.266	2.223	2.178	2.131
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.247	2.204	2.160	2.114	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.194	2.151	2.106	2.059	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.148	2.104	2.058	2.011	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.107	2.063	2.017	1.968	1.917
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.071	2.026	1.980	1.930	1.878
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.039	1.994	1.946	1.896	1.843
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.010	1.965	1.916	1.866	1.812
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	1.984	1.938	1.889	1.838	1.783
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	1.961	1.914	1.865	1.813	1.757
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.939	1.892	1.842	1.790	1.733
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.841	1.792	1.740	1.683	1.622
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.744	1.693	1.637	1.577	1.509
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.649	1.594	1.534	1.467	1.389
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.554	1.495	1.429	1.352	1.254
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.459	1.394	1.318	1.221	1.000

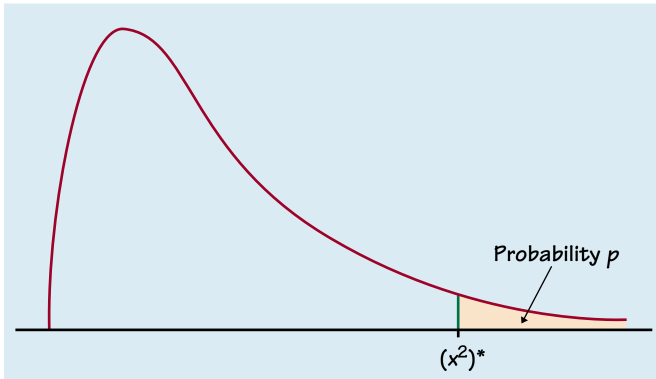


Table entry for p is the point $(X^2)^*$ with probability p lying above it.

TABLE F X^2 distribution critical values

	Tail probability p											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.32	1.64	2.07	2.71	3.84	5.02	5.41	6.63	7.88	9.14	10.83	12.12
2	2.77	3.22	3.79	4.61	5.99	7.38	7.82	9.21	10.60	11.98	13.82	15.20
3	4.11	4.64	5.32	6.25	7.81	9.35	9.84	11.34	12.84	14.32	16.27	17.73
4	5.39	5.99	6.74	7.78	9.49	11.14	11.67	13.28	14.86	16.42	18.47	20.00
5	6.63	7.29	8.12	9.24	11.07	12.83	13.39	15.09	16.75	18.39	20.51	22.11
6	7.84	8.56	9.45	10.64	12.59	14.45	15.03	16.81	18.55	20.25	22.46	24.10
7	9.04	9.80	10.75	12.02	14.07	16.01	16.62	18.48	20.28	22.04	24.32	26.02
8	10.22	11.03	12.03	13.36	15.51	17.53	18.17	20.09	21.95	23.77	26.12	27.87
9	11.39	12.24	13.29	14.68	16.92	19.02	19.68	21.67	23.59	25.46	27.88	29.67
10	12.55	13.44	14.53	15.99	18.31	20.48	21.16	23.21	25.19	27.11	29.59	31.42
11	13.70	14.63	15.77	17.28	19.68	21.92	22.62	24.72	26.76	28.73	31.26	33.14
12	14.85	15.81	16.99	18.55	21.03	23.34	24.05	26.22	28.30	30.32	32.91	34.82
13	15.98	16.98	18.20	19.81	22.36	24.74	25.47	27.69	29.82	31.88	34.53	36.48
14	17.12	18.15	19.41	21.06	23.68	26.12	26.87	29.14	31.32	33.43	36.12	38.11
15	18.25	19.31	20.60	22.31	25.00	27.49	28.26	30.58	32.80	34.95	37.70	39.72
16	19.37	20.47	21.79	23.54	26.30	28.85	29.63	32.00	34.27	36.46	39.25	41.31
17	20.49	21.61	22.98	24.77	27.59	30.19	31.00	33.41	35.72	37.95	40.79	42.88
18	21.60	22.76	24.16	25.99	28.87	31.53	32.35	34.81	37.16	39.42	42.31	44.43
19	22.72	23.90	25.33	27.20	30.14	32.85	33.69	36.19	38.58	40.88	43.82	45.97
20	23.83	25.04	26.50	28.41	31.41	34.17	35.02	37.57	40.00	42.34	45.31	47.50
21	24.93	26.17	27.66	29.62	32.67	35.48	36.34	38.93	41.40	43.78	46.80	49.01
22	26.04	27.30	28.82	30.81	33.92	36.78	37.66	40.29	42.80	45.20	48.27	50.51
23	27.14	28.43	29.98	32.01	35.17	38.08	38.97	41.64	44.18	46.62	49.73	52.00
24	28.24	29.55	31.13	33.20	36.42	39.36	40.27	42.98	45.56	48.03	51.18	53.48
25	29.34	30.68	32.28	34.38	37.65	40.65	41.57	44.31	46.93	49.44	52.62	54.95
26	30.43	31.79	33.43	35.56	38.89	41.92	42.86	45.64	48.29	50.83	54.05	56.41
27	31.53	32.91	34.57	36.74	40.11	43.19	44.14	46.96	49.64	52.22	55.48	57.86
28	32.62	34.03	35.71	37.92	41.34	44.46	45.42	48.28	50.99	53.59	56.89	59.30
29	33.71	35.14	36.85	39.09	42.56	45.72	46.69	49.59	52.34	54.97	58.30	60.73
30	34.80	36.25	37.99	40.26	43.77	46.98	47.96	50.89	53.67	56.33	59.70	62.16
40	45.62	47.27	49.24	51.81	55.76	59.34	60.44	63.69	66.77	69.70	73.40	76.09
50	56.33	58.16	60.35	63.17	67.50	71.42	72.61	76.15	79.49	82.66	86.66	89.56
60	66.98	68.97	71.34	74.40	79.08	83.30	84.58	88.38	91.95	95.34	99.61	102.7
80	88.13	90.41	93.11	96.58	101.9	106.6	108.1	112.3	116.3	120.1	124.8	128.3
100	109.1	111.7	114.7	118.5	124.3	129.6	131.1	135.8	140.2	144.3	149.4	153.2